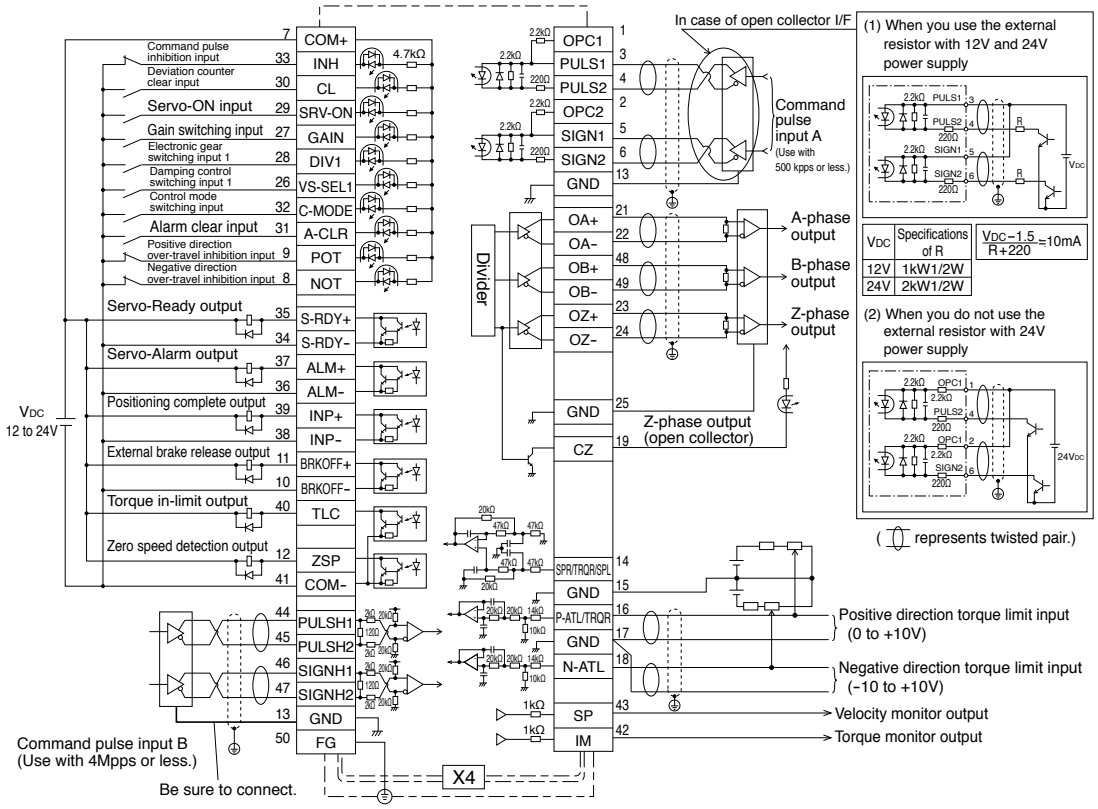
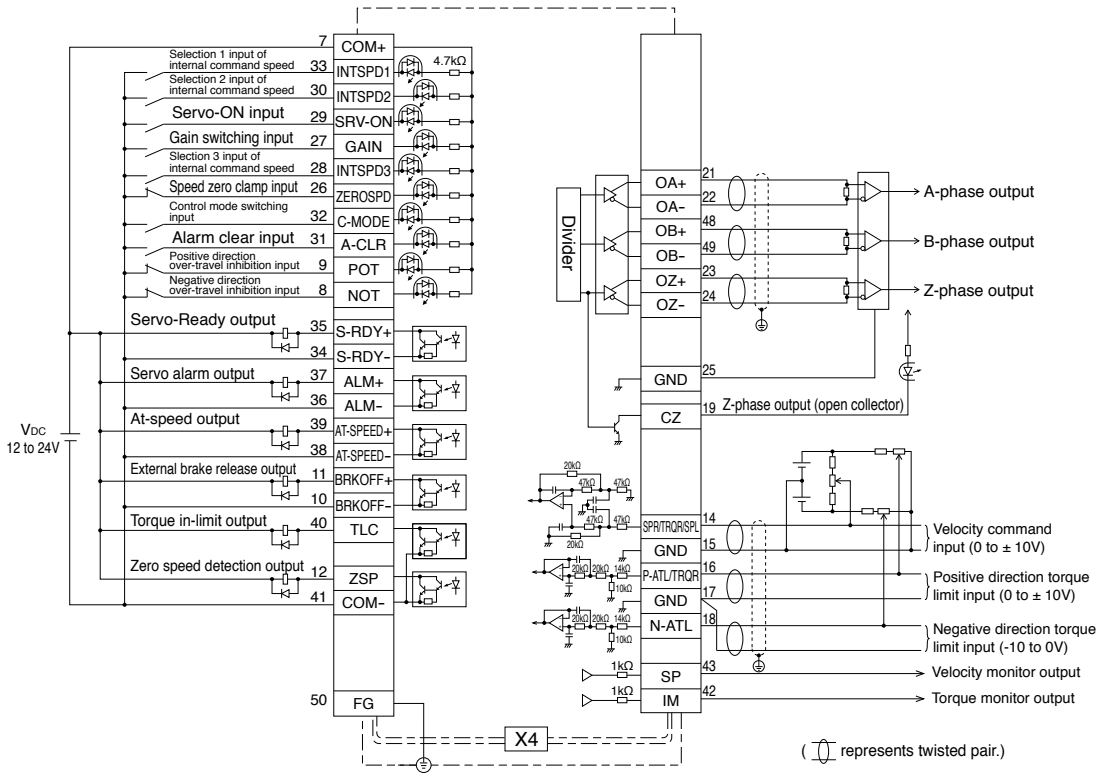
3. Wiring Diagram to the connector, X4

Example of control mode specific wiring

Wiring Example of Position Control Mode

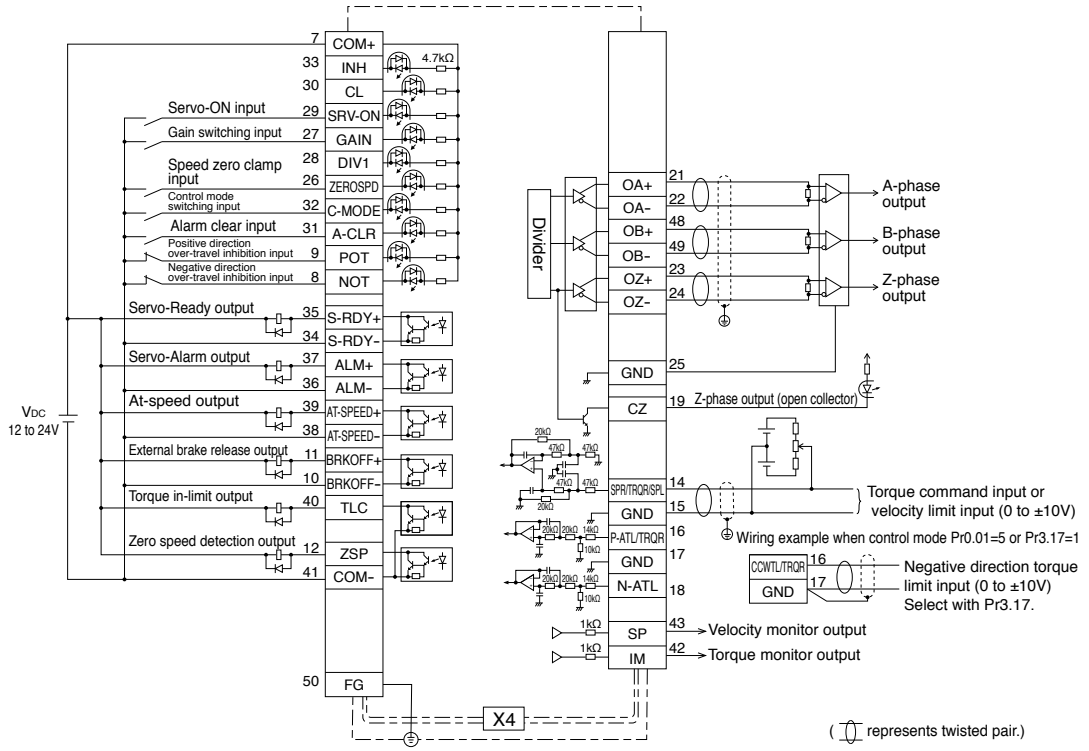


Wiring Example of Velocity Control Mode

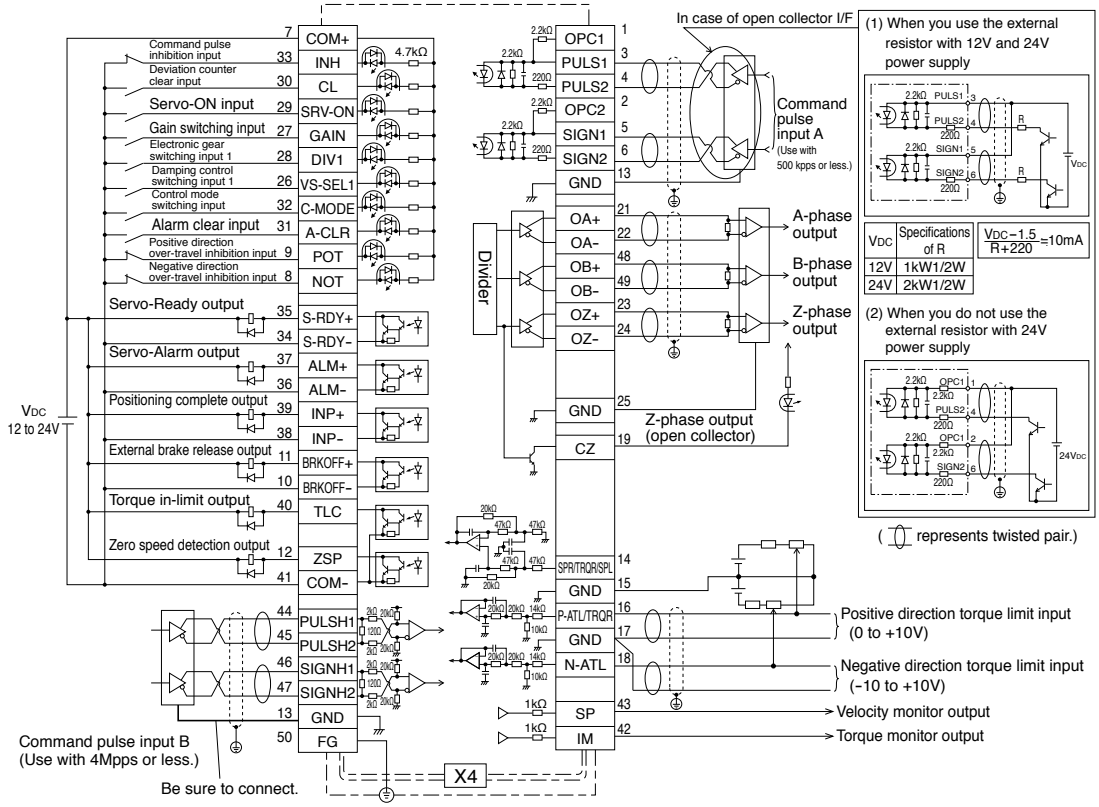


Note The functions of the following pin can be changed using parameters. (Refer to P.4-32)
Input(Position): 8, 9, 26, 27, 28, 29, 31, 32 Input(Velocity): 8, 9, 26, 27, 28, 29, 30, 31, 32, 33
Output: 10-11, 12, 34-35, 36-37, 38-39, 40
* Pins in the figure above represent default parameter values.

Wiring Example of Torque Control Mode



Wiring Example of Full-closed Control Mode

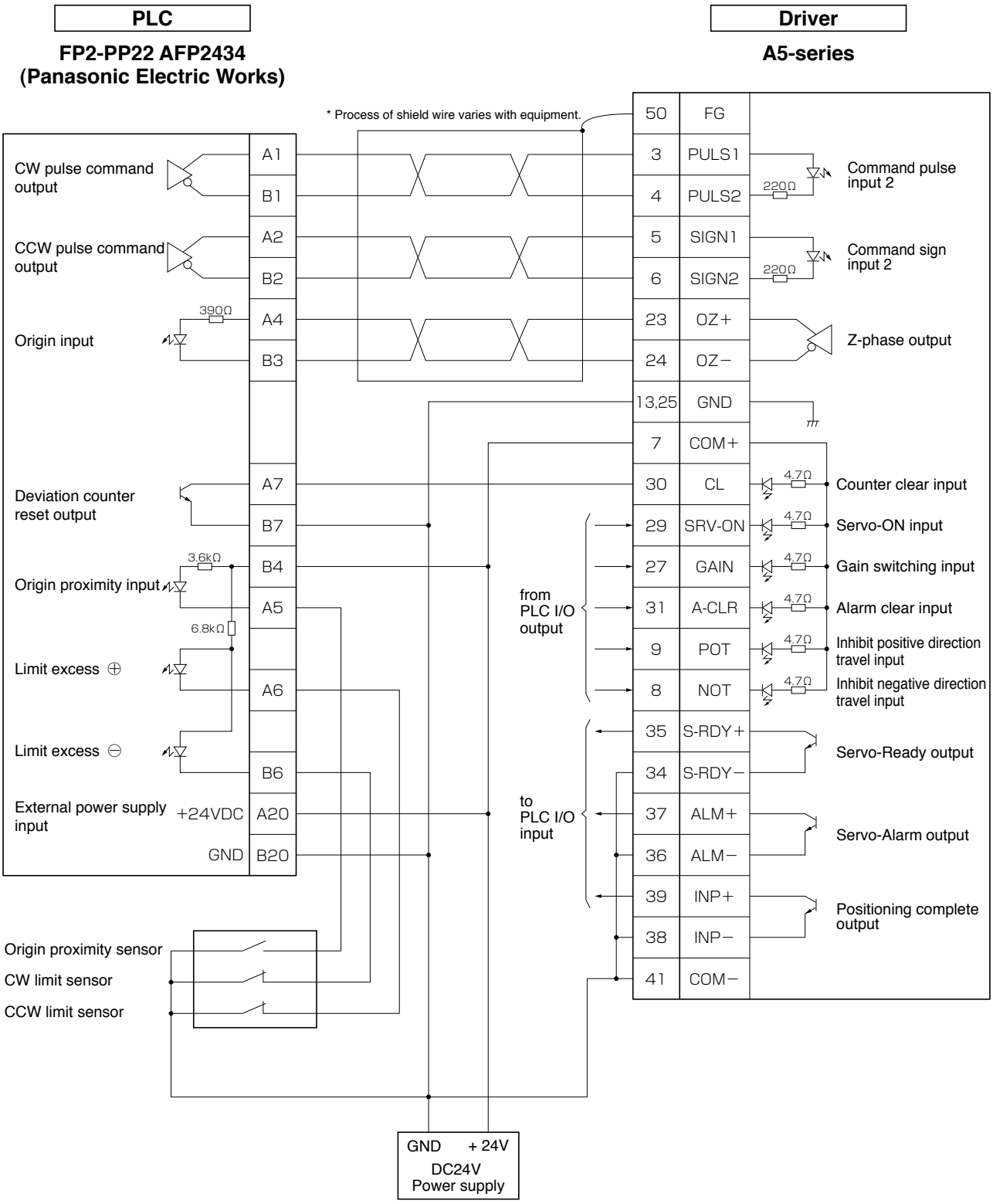


Note The functions of the following pin can be changed using parameters. (Refer to P.4-32)
Input(Torque): 8, 9, 26, 27, 28, 29, 30, 31, 32, 33 Input(Full-closed): 8, 9, 26, 27, 28, 29, 31, 32
Output: 10-11, 12, 34-35, 36-37, 38-39, 40
* Pins in the figure above represent default parameter values.

3. Wiring Diagram to the connector, X4

Connecting Example to Host Controller

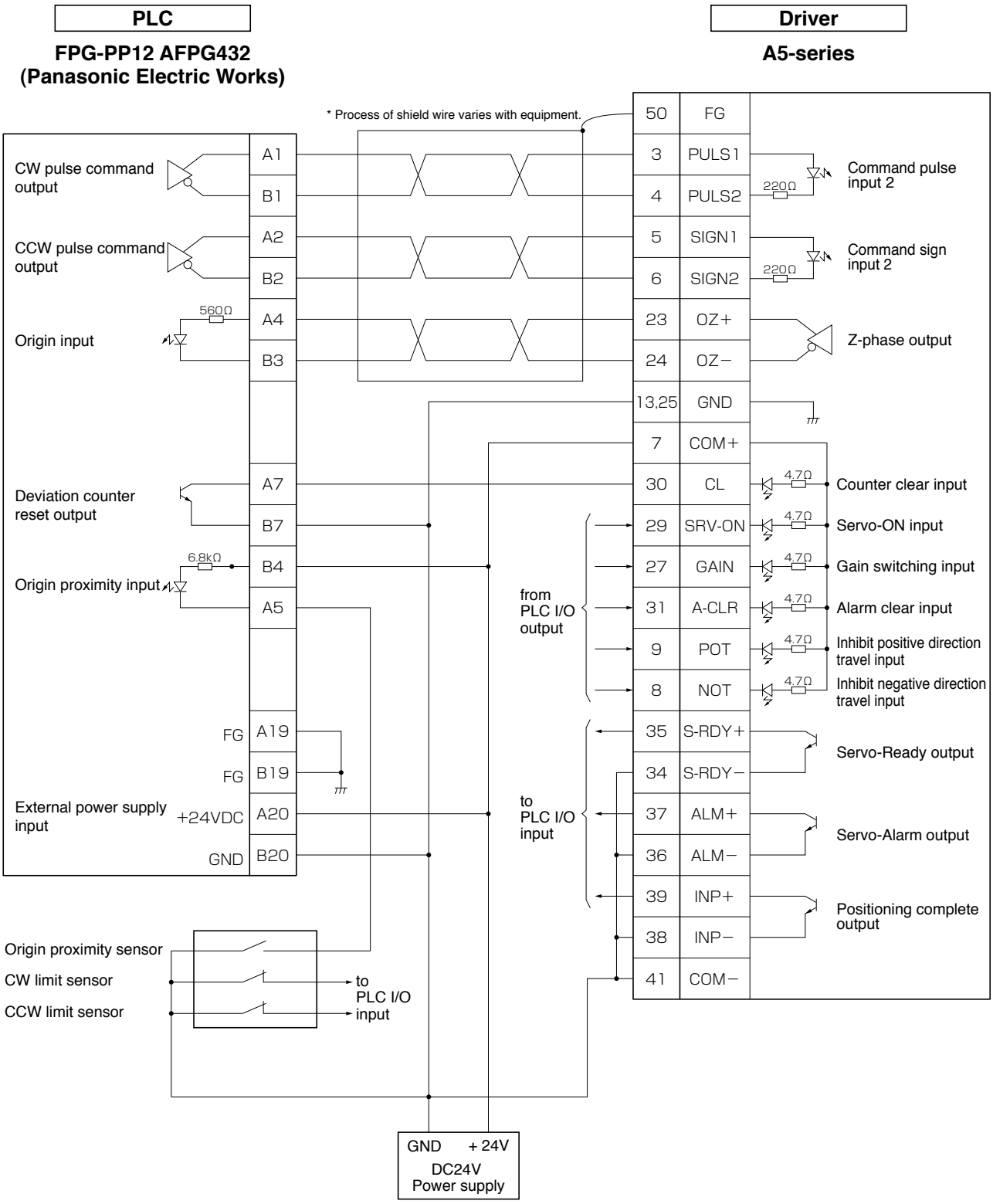
Connection between MINAS A5 and FP2-PP22 AFP2434 (Panasonic Electric Works)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

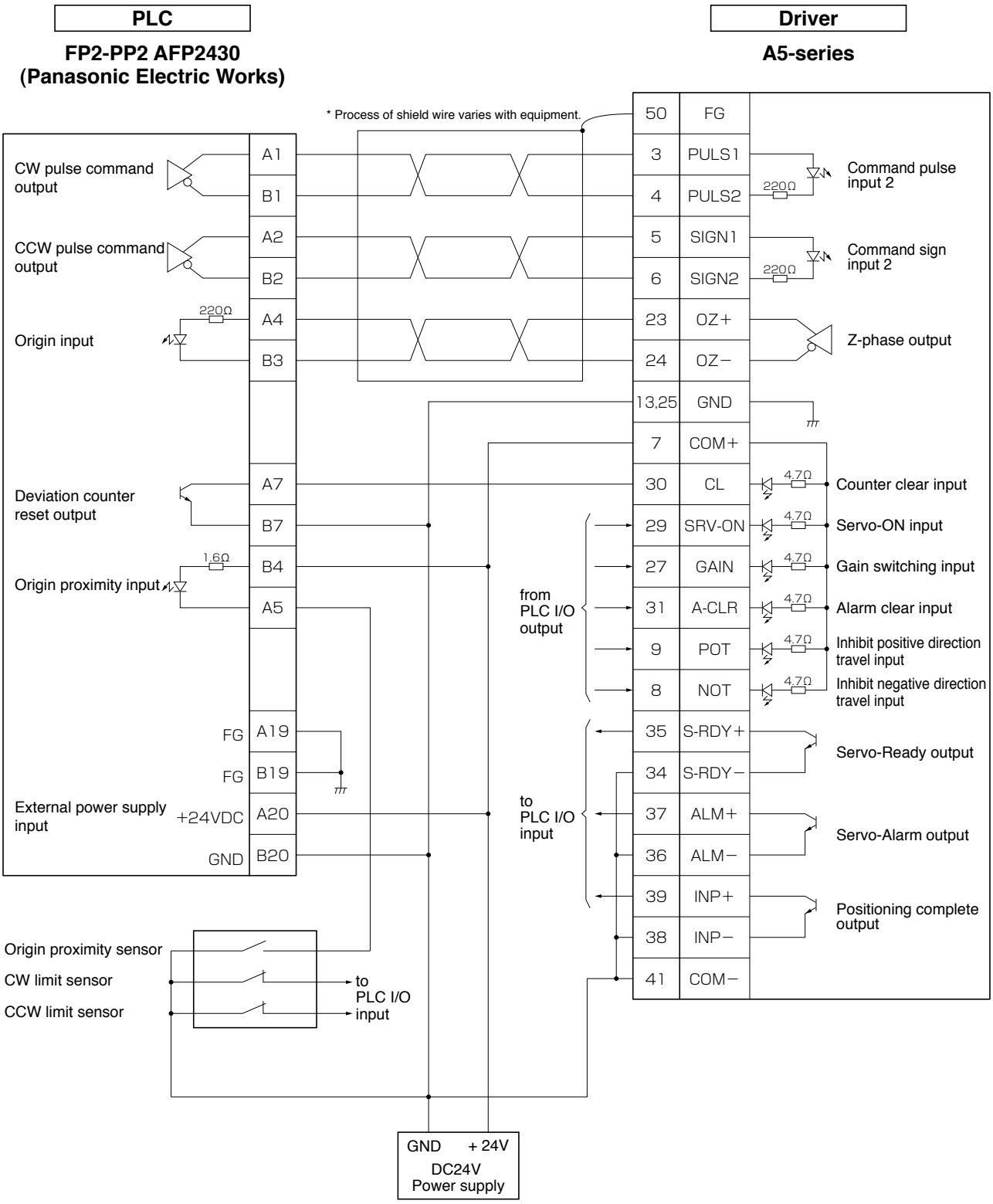
Connection between MINAS A5 and FPG-PP12 AFPG432 (Panasonic Electric Works)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

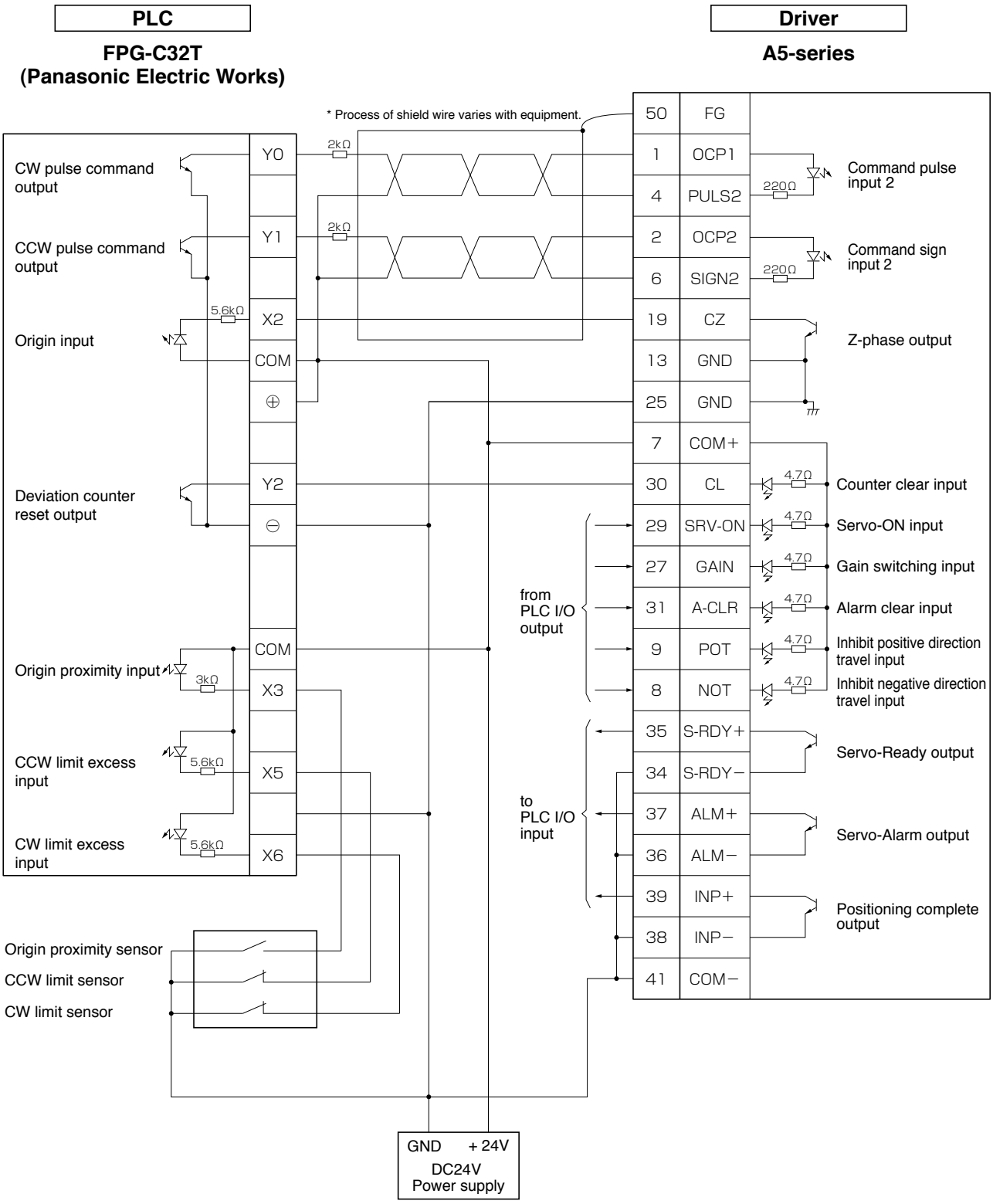
Connection between MINAS A5 and FP2-PP22 AFP2434 (Panasonic Electric Works)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

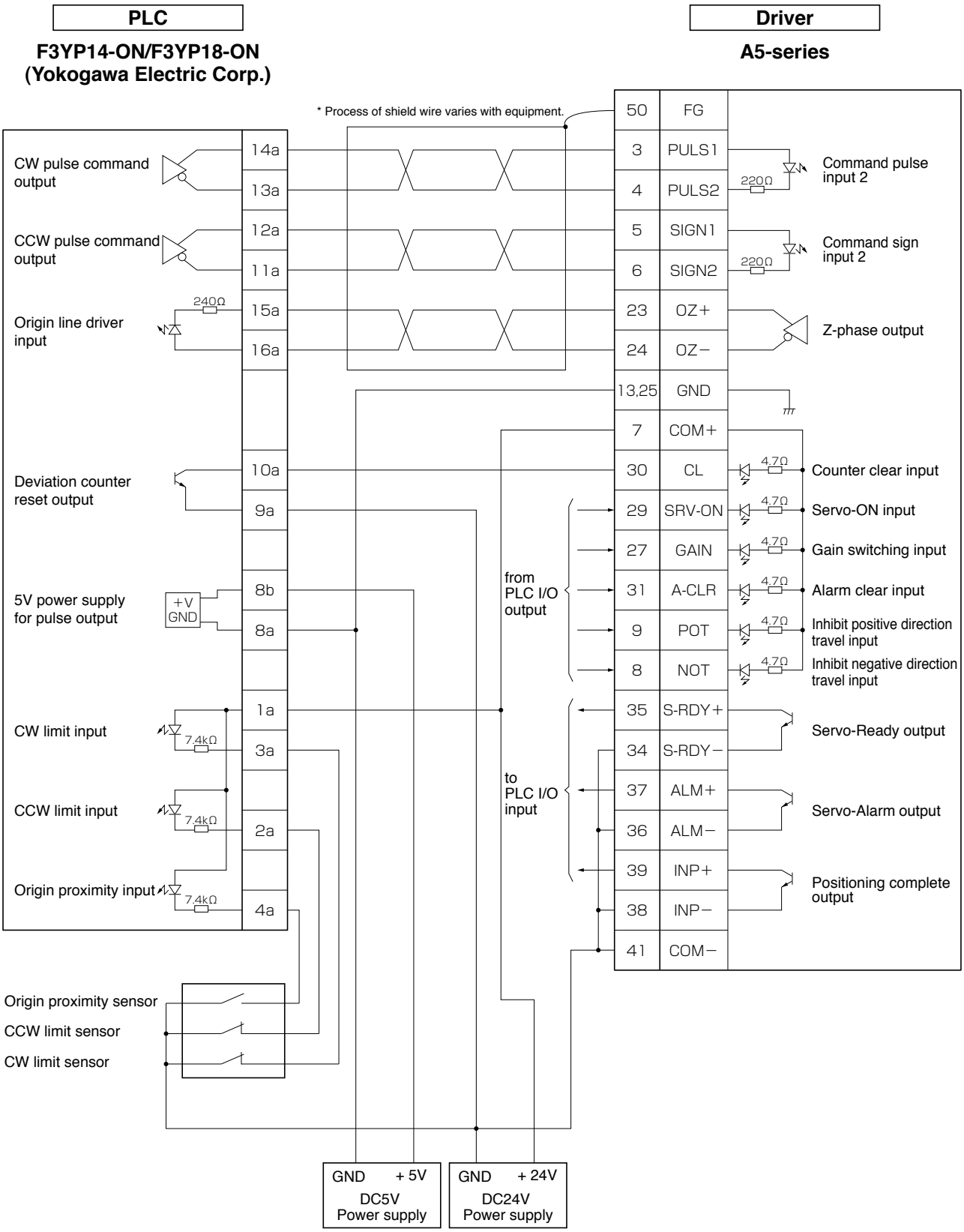
Connection between MINAS A5 and FPG-C32T (Panasonic Electric Works)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

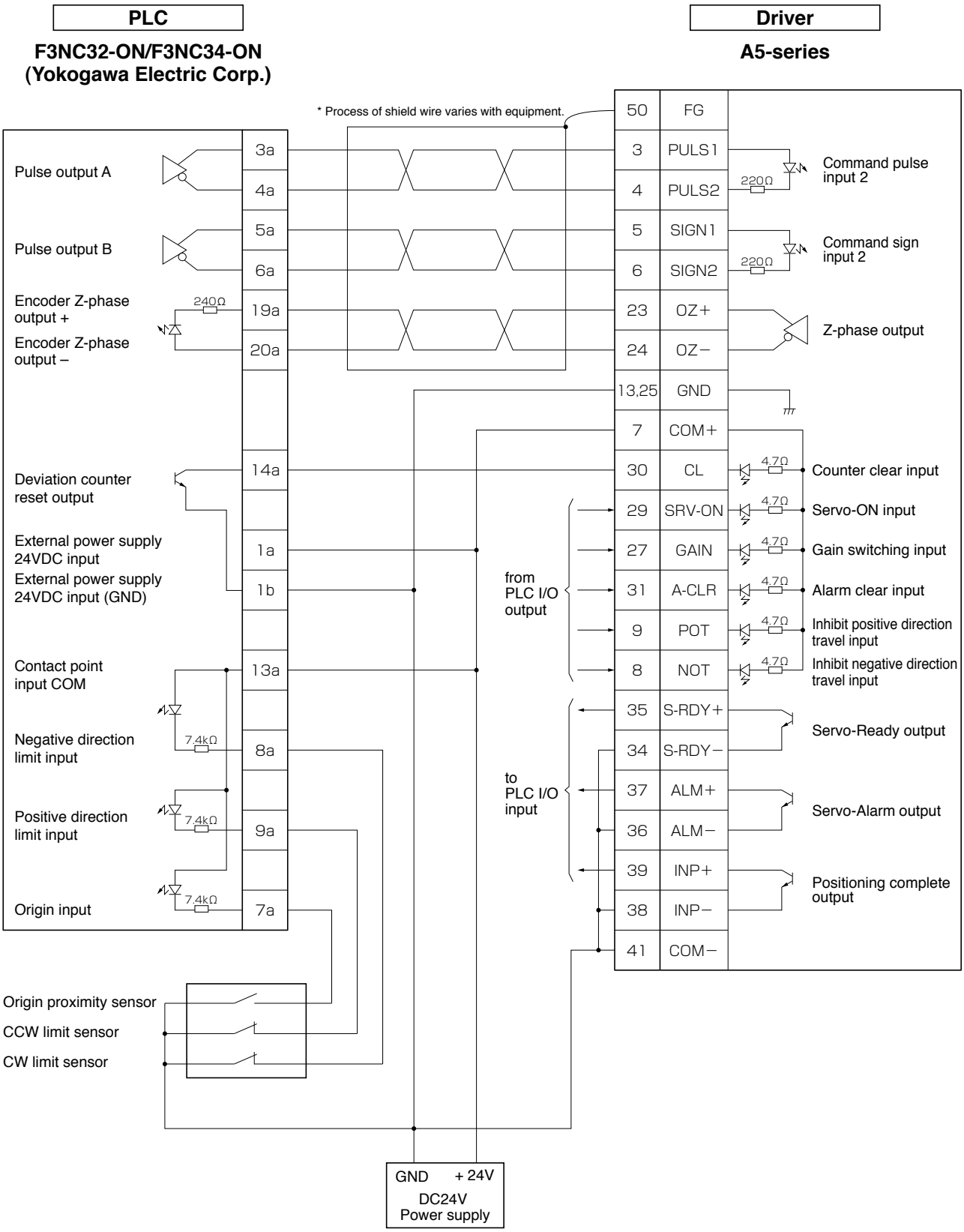
Connection between MINAS A5 and F3YP14-ON/F3YP18-ON (Yokogawa Electric Corp.)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

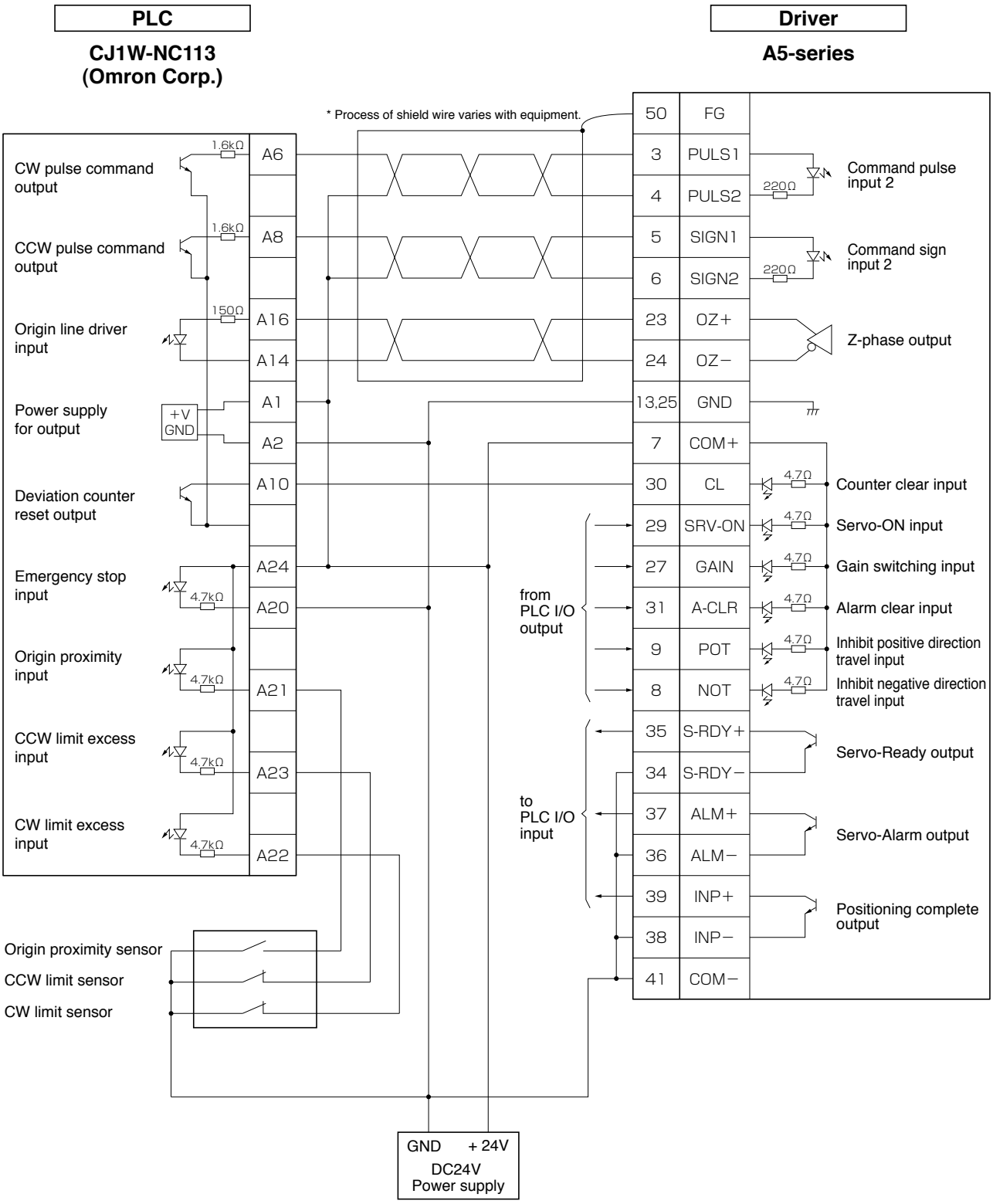
Connection between MINAS A5 and F3NC32-ON/F3NC34-ON (Yokogawa Electric Corp.)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

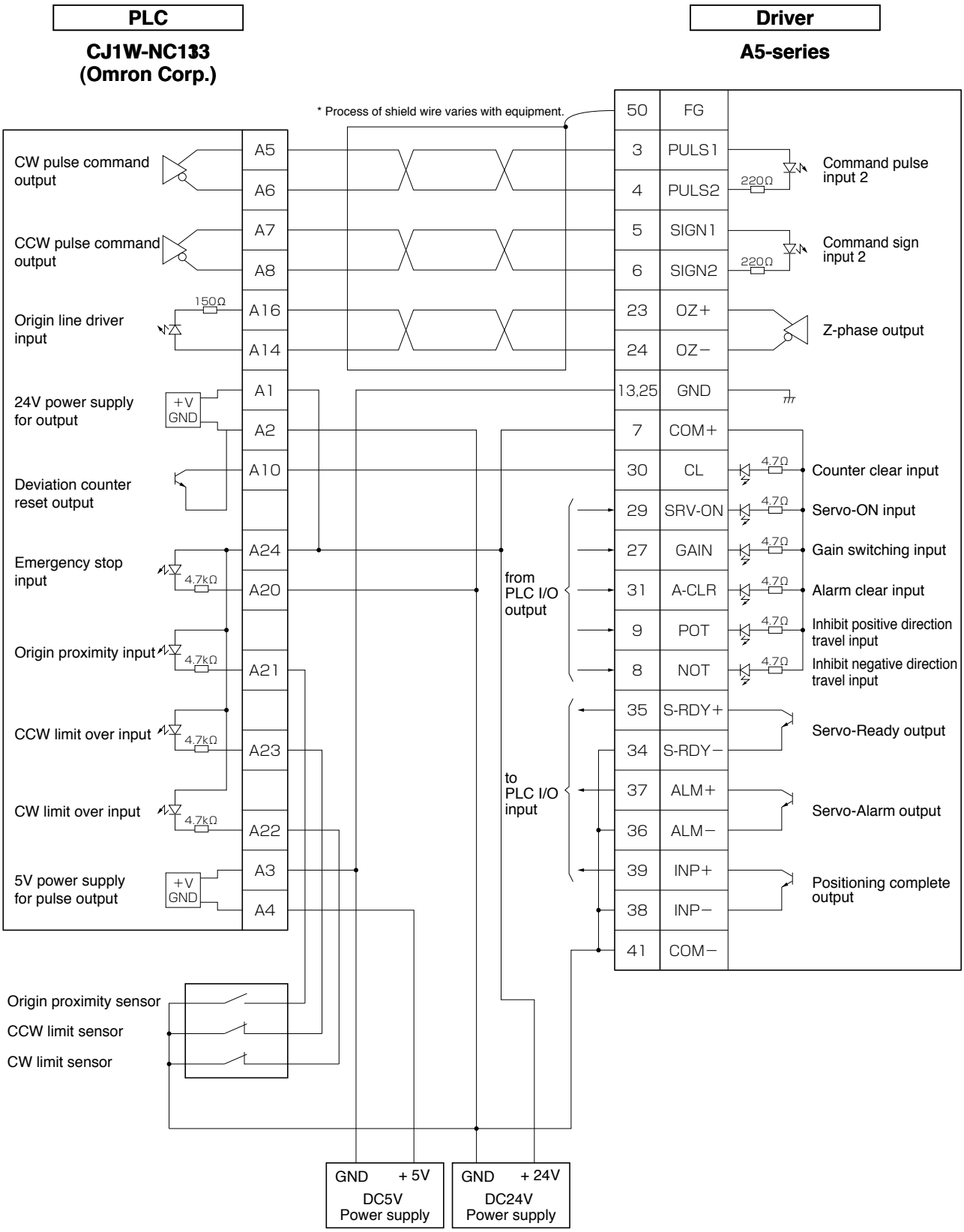
Connection between MINAS A5 and CJ1W-NC113 (Omron Corp.)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

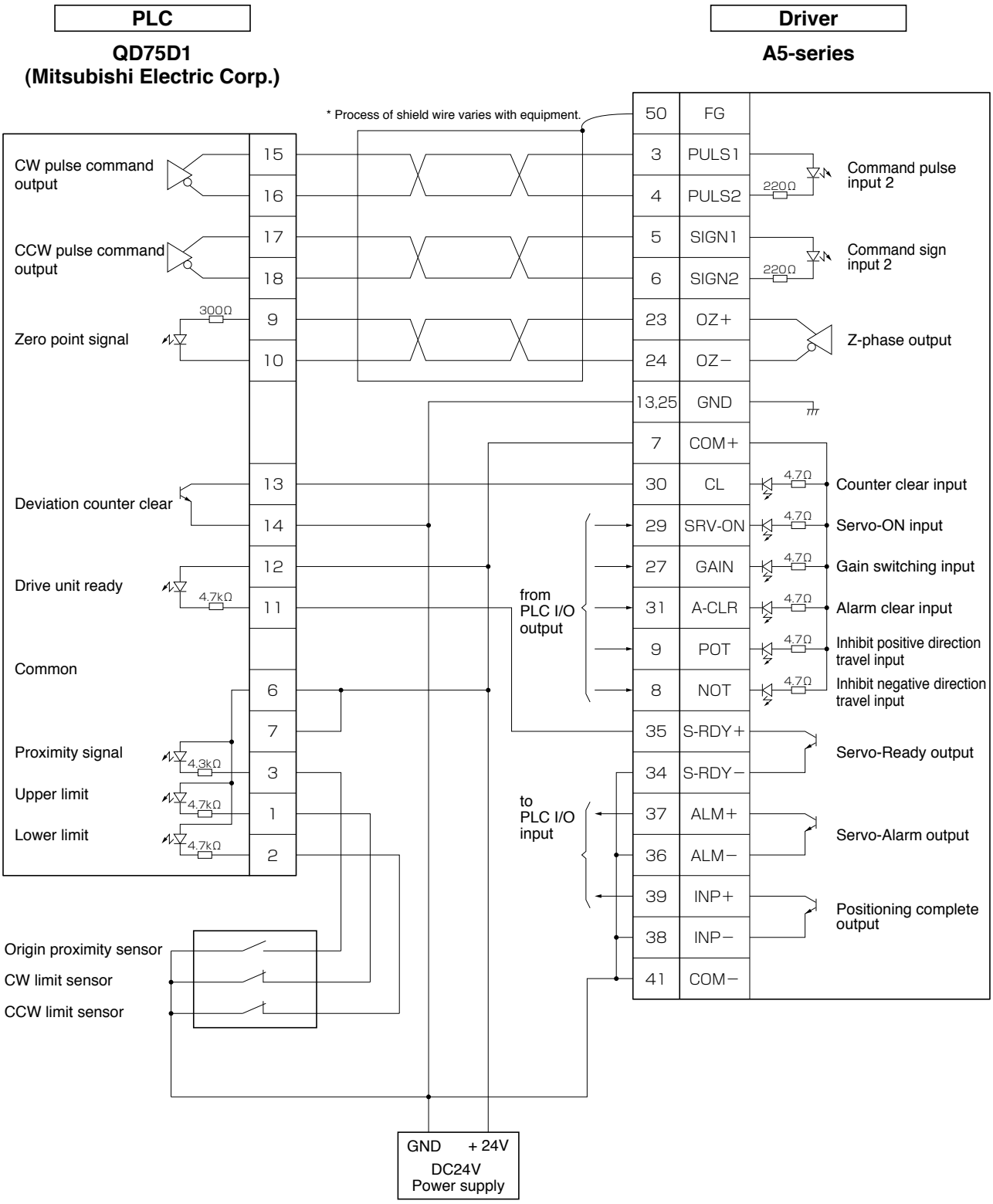
Connection between MINAS A5 and CJ1W-NC133 (Omron Corp.)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

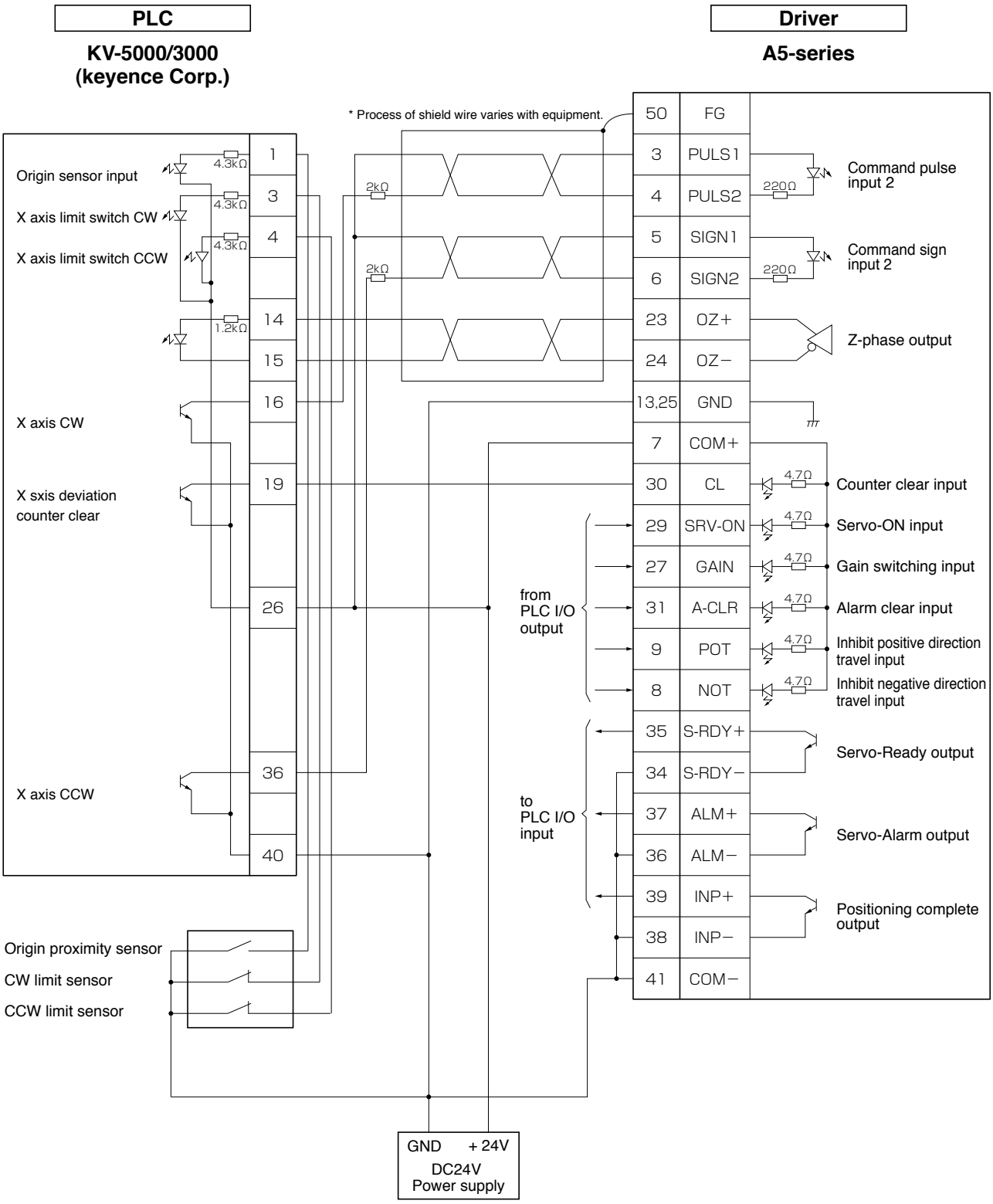
Connection between MINAS A5 and QD75D1 (Mitsubishi Electric Corp.)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

Connection between MINAS A5 and KV-5000/3000 (keyence Corp.)



Note represents twisted pair wire.

Related page • P.3-30 “Inputs and outputs on connector X4”

4. Inputs and outputs on connector X4

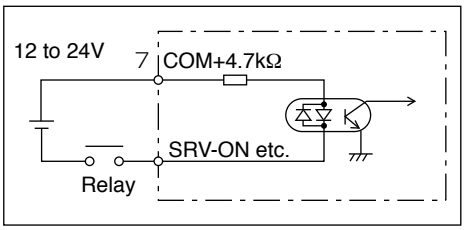
Interface Circuit (Input)

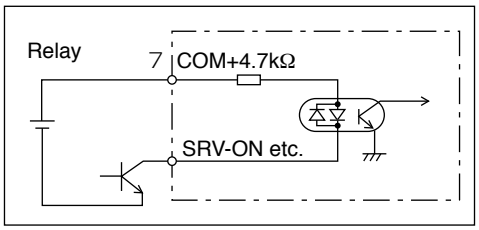
Input Circuit

SI — Connection to sequence input signals

Related control mode: P S T F

- Connect to contacts of switches and relays, or open collector output transistors.
- When you use contact inputs, use the switches and relays for micro current to avoid contact failure.
- Make the lower limit voltage of the power supply (12 to 24V) as 11.4V or more in order to secure the primary current for photo-couplers.





PI1 — Connection to sequence input signals (Pulse train interface)

Related control mode: P S T F

(1) Line driver I/F (Input pulse frequency : max. 500kpps)

- This signal transmission method has better noise immunity. We recommend this to secure the signal transmission.

(2) Open collector I/F (Input pulse frequency : max. 200kpps)

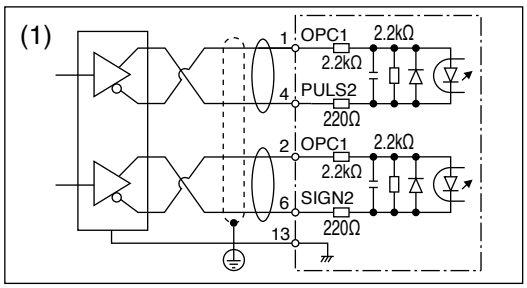
- The method which uses an external control signal power supply (VDC)
- Current regulating resistor (R) corresponding to V_{DC} is required in this case.
- Connect the specified resister as below.
- (R) should be placed close to the driver for effective noise reduction.

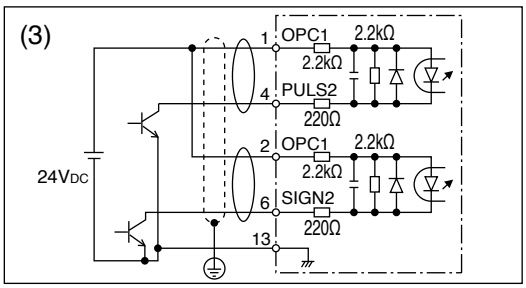
VDC	Specifications
12V	1kΩ1/2W
24V	2kΩ1/2W

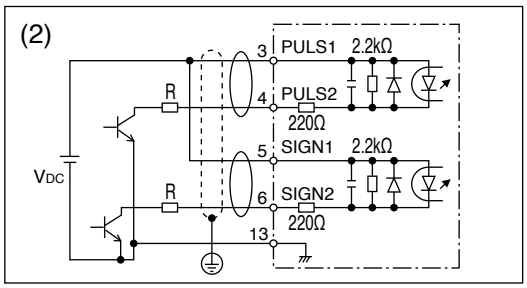
$$\frac{V_{DC} - 1.5}{R + 220} \fallingdotseq 10mA$$

(3) Open collector I/F (Input pulse frequency : max. 200kpps)

- Connecting diagram when a current regulating resistor is not used with 24V power supply.







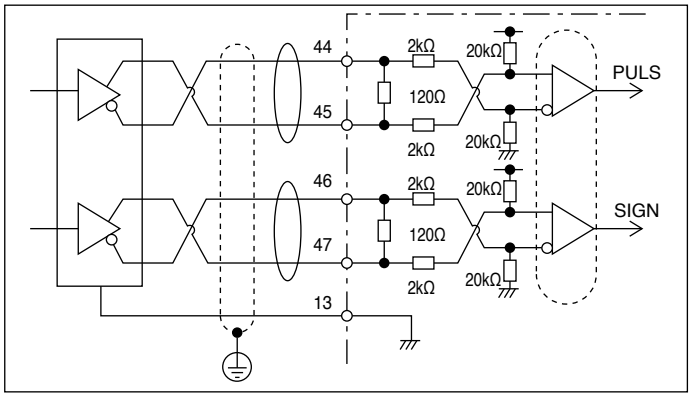
Max.input voltage :
DC24V, Rated current : 10mA

represents twisted pair.

PI2	Connection to sequence input signals (Pulse train interface exclusive to line driver)	Related control mode	P	S	T	F

Line driver I/F (Input pulse frequency : max. 4Mpps)

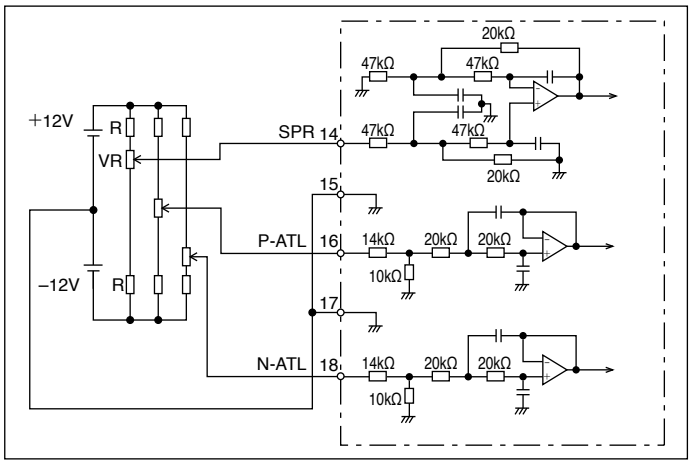
- This signal transmission method has better noise immunity.
  We recommend this to secure the signal transmission when line driver I/F is used.



represents twisted pair.

AI	Analog command input	Related control mode	P	S	T	F

- The analog command input goes through 3 routes, SPR (Pin-14), P-ATL (Pin-16) and N-ATL (Pin-18).
- Max. permissible input voltage to each input is ±10V.
  For input impedance of each input, refer to the right Fig.
- When you compose a simple command circuit using variable resistor(VR) and register R, connect as the right Fig. shows. When the variable range of each input is made as –10V to +10V, use VR with 2kW, B-characteristics, 1/2W or larger, R with 200W, 1/2W or larger.
- A/D converter resolution of each command input is as follows.
  (1)ADC1 : 16 bit (SPR/TRQR)
  (2)ADC2 : 12 bit (P-ATL and N-ATL)



represents twisted pair.

4. Inputs and outputs on connector X4

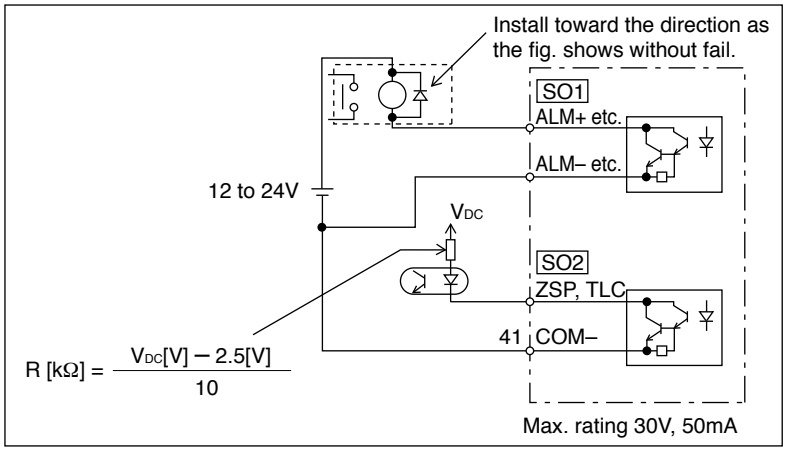
Interface Circuit (Output)

Output Circuit

SO1/SO2 Sequence output circuit

Related control mode: P S T F

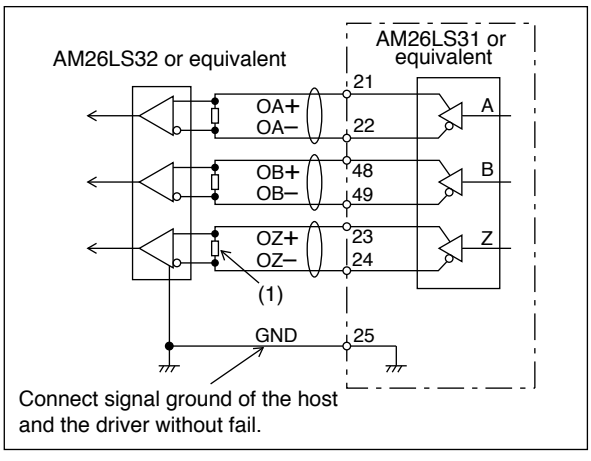
- The output circuit is composed of open collector transistor outputs in the Darlington connection, and connect to relays or photo-couplers.
- There exists collector to emitter voltage, V_{CE} (SAT) of approx. 1V at transistor-ON, due to the Darlington connection of the output or. Note that normal TTL IC cannot be directly connected since it does not meet VIL.
- There are two types of output, one which emitter side of the output transistor is independent and is connectable individually, and the one which is common to – side of the control power supply (COM–).
- If a recommended primary current value of the photo-coupler is 10mA, decide the resistor value using the formula of the right Fig.



PO1 Line driver (Differential output) output

Related control mode: P S T F

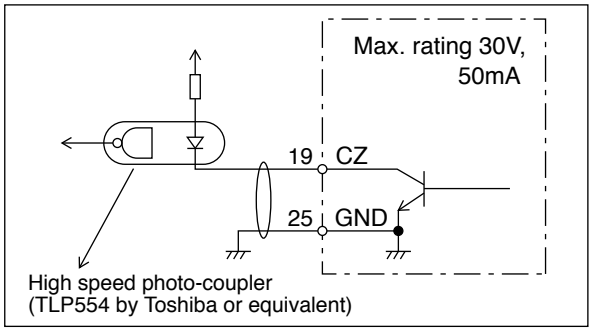
- Feeds out the divided encoder outputs (A, B and Z-phase) in differential through each line driver.
- At the host side, receive these in line receiver. Install a terminal resistor (approx. 330Ω) (figure below (1))between line receiver inputs without fail.
- These outputs are not insulated.



represents twisted pair.

PO2	Open collector output	Related control mode	P	S	T	F

- Feeds out the Z-phase signal among the encoder signals in open collector. This output is not insulated.
- Receive this output with high-speed photo couplers at the host side, since the pulse width of the Z-phase signal is narrow.



represents twisted pair.

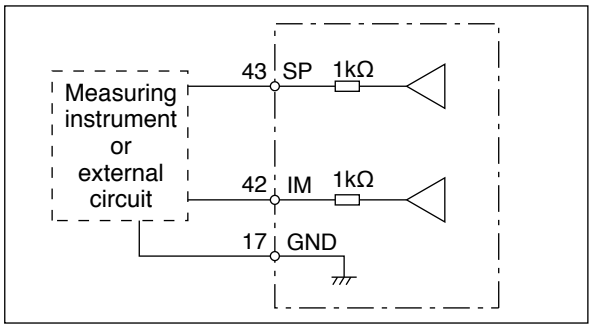
AO	Analog monitor output	Related control mode	P	S	T	F

- There are two outputs, the speed monitor signal output (SP) and the torque monitor signal output (IM)
- Output signal width is ±10V.
- The output impedance is 1kΩ. Pay an attention to the input impedance of the measuring instrument or the external circuit to be connected.

<Resolution>

(1) Speed monitor output (SP)
With a setup of 6V/3000r/min, the resolution converted to speed is 8r/min/16mV.

(2) Torque monitor output (IM)
With a relation of 3V/rated torque (100%), the resolution converted to torque is 0.4%/12mV.



4. Inputs and outputs on connector X4

Input Signal and Pin No.

Input Signals (common) and Their Functions

Pin No.	7	Title of signal	Power supply for control signal (+)	Related control mode	P	S	T	F
		Symbol	COM+	I/F circuit	—			

- Connect + of the external DC power supply (12 to 24V).
- Use the power supply voltage of 12V ± 5% – 24V ± 5%

Pin No.	41	Title of signal	Power supply for control signal (-)	Related control mode	P	S	T	F
		Symbol	COM–	I/F circuit	—			

- Connect – of the external DC power supply (12 to 24V).
- The power capacity varies depending on a composition of I/O circuit. 0.5A or more is recommended.

Input Signals (Pulse Train) and Their Functions

You can select appropriate interface out of two kinds, depending on the command pulse specifications.

• Pulse train interface exclusive for line driver

Pin No.		Title of signal		Related control mode	P	S	T	F
	44 45	Command pulse input 1						
		Symbol	Pin No.44: **PULSH1** Pin No.45: **PULSH2**	I/F circuit	PI2	3-31 page		

Pin No.		Title of signal		Related control mode	P	S	T	F
	46 47	Command pulse sign input 1						
		Symbol	Pin No.46: **SIGNH2** Pin No.47: **SIGNH2**	I/F circuit	PI2	3-31 page		

- Input terminal for position command pulse. You can select by setting up Pr0.05 (Selection of command pulse input) to 1.
- This input becomes invalid at such control mode as velocity control or torque control, where no position command is required.
- Permissible max. input frequency is 4Mpps.
- You can select up to 6 command pulse input formats with Pr0.06 (Setup of command pulse rotational direction) and Pr0.07 (Setup of command pulse input mode). For details, refer to the table next page, "Command pulse input format".

• Pulse train interface (supports both line driver and open collector)

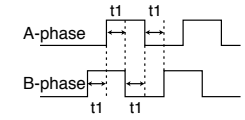
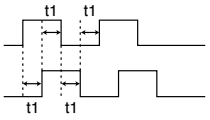
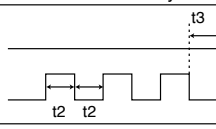
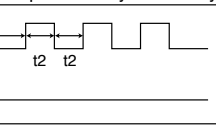
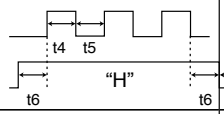
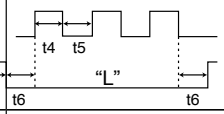
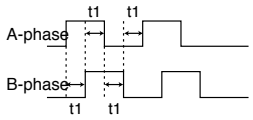
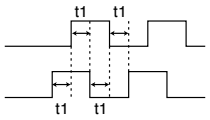
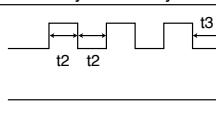
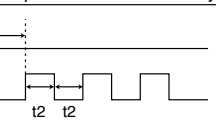
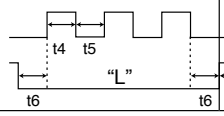
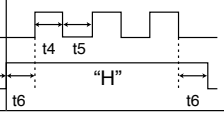
Pin No.		Title of signal		Related control mode	P	S	T	F
	1 3 4	Command pulse input 2						
		Symbol	Pin No.1: **OPC1** Pin No.3: **PULS1** Pin No.4: **PULS2**	I/F circuit	PI1	3-30 page		

Pin No.		Title of signal		Related control mode	P	S	T	F
	2 5 6	Command pulse sign input 2						
		Symbol	Pin No.2: **OPC1** Pin No.5: **SIGN1** Pin No.6: **SIGN2**	I/F circuit	PI1	3-30 page		

- Input terminal for the position command. You can select by setting up Pr0.05 (Selection of command pulse input) to 0.
- This input becomes invalid at such control mode as the velocity control or torque control, where no position command is required.
- Permissible max. input frequency is 500kpps at line driver input and 200kpps at open collector input.
- You can select up to 6 command pulse input formats with Pr0.06 (Setup of command pulse rotational direction) and Pr0.07 (Setup of command pulse input mode). For details, refer to the table next page, "Command pulse input format".

Related page
- P.3-30 “Inputs and outputs on connector X4”
- P.4-4 “Details of parameter”

• **Input format command pulse**

Pr0.06 setup value (Command pulse rotational direction setup)	Pr0.07 setup value (Command pulse input mode setup)	Command pulse format	Signal title	Positive direction command	Negative direction command
0	0 or 2	90˚ phase difference 2-phase pulse (A + B-phase)	PULS SIGN	 B-phase advances to A by 90˚.	 B-phase delays from A by 90˚.
	1	Positive direction pulse train + Negative direction pulse train	PULS SIGN		
	3	pulse train + Signal	PULS SIGN		
1	0 or 2	90˚ phase difference 2-phase pulse (A + B-phase)	PULS SIGN	 B-phase delays from A by 90˚.	 B-phase advances to A by 90˚.
	1	Positive direction pulse train + Negative direction pulse train	PULS SIGN		
	3	pulse train + Signal	PULS SIGN		

• PULS and SIGN represents the outputs of pulse train in put circuit. Refer to the fig. of P.3-30, "Input Circuit".
• In case of negative direction pulse train + positive direction pulse train and pulse train + sign, pulse train will be cap tured at the rising edge.
• In case of 2-phase pulse, pulse train will be captured at each edge.

• **Permissible max. input frequency, and min. necessary time width of command pulse input signal.**

Input I/F of PULS/SIGN signal		Permissible max. input frequency	Min. necessary time width (μs)					
			t_1	t_2	t_3	t_4	t_5	t_6
Pulse train interface exclusive to line driver		4Mpps	0.25	0.125	0.125	0.125	0.125	0.125
Pulse train interface	Line driver interface	500kpps	2	1	1	1	1	1
	Open collector interface	200kpps	5	2.5	2.5	2.5	2.5	2.5

Make the rising/falling time of the command pulse input signal to 0.1μs or smaller.

Related page • P.4-7 "Details of Parameter"

Control Input

Control signal having the desired function can be applied to any input pin of I/F connector. The logic can be changed.

• Default assignment

				Applicable parameter	Default parameter setting ( ): decimal notation	Default Setup					
						Position/ Full-closed control		Verocity control		Torque control	
						Signal	Logic *1	Signal	Logic *1	Signal	Logic *1
Pin No.	8	Title of signal	SI1 input	Pr4.00	00828282h (8553090)	NOT	b-contact	NOT	b-contact	NOT	b-contact
		Symbol	SI1								
Pin No.	9	Title of signal	SI2 input	Pr4.01	00818181h (8487297)	POT	b-contact	POT	b-contact	POT	b-contact
		Symbol	SI2								
Pin No.	26	Title of signal	SI3 input	Pr4.02	0091918Ah (9539850)	VS-SEL1	b-contact	ZEROSPD	b-contact	ZEROSPD	b-contact
		Symbol	SI3								
Pin No.	27	Title of signal	SI4 input	Pr4.03	00060606h (394758)	GAIN	a-contact	GAIN	a-contact	GAIN	a-contact
		Symbol	SI4								
Pin No.	28	Title of signal	SI5 input	Pr4.04	0000100Ch (4108)	DIV1	b-contact	INTSPD3	b-contact	—	—
		Symbol	SI5								
Pin No.	29	Title of signal	SI6 input	Pr4.05	00030303h (197379)	SRV-ON	a-contact	SRV-ON	a-contact	SRV-ON	a-contact
		Symbol	SI6								
Pin No.	30	Title of signal	SI7 input	Pr4.06	00000f07h (3847)	CL	a-contact	INTSPD2	a-contact	—	—
		Symbol	SI7								
Pin No.	31	Title of signal	SI8 input	Pr4.07	00040404h (263172)	A-CLR	a-contact	A-CLR	a-contact	A-CLR	a-contact
		Symbol	SI8								
Pin No.	32	Title of signal	SI9 input	Pr4.08	00050505h (328965)	C-MODE	a-contact	C-MODE	a-contact	C-MODE	a-contact
		Symbol	SI9								
Pin No.	33	Title of signal	SI10 input	Pr4.09	00000E88h (3720)	INH	b-contact	INTSPD1	b-contact	—	—
		Symbol	SI10								

• The function depends on parameter setup. Refer to P.4-32, 4-33.
Refer to the next section “Function assignable to general purpose input”.

Note *1 Operation of a-contact and b-contact:

a-contact: Input signal disconnected from COM– function disabled (OFF state)
Input signal connected to COM– function enabled (ON state)

b-contact: Input signal disconnected from COM– function enabled (ON state)
Input signal connected to COM function disabled (OFF state)

[—]: No function assigned

Function allocatable to control input

Title of signal	Servo-ON input			Related control mode	P	S	T	F
Symbol	SRV-ON	Default assignment	SI6	I/F circuit	SI	3-30 page		

- This signal turns on/off the servo (motor).

Title of signal	Positive direction over-travel inhibition input			Related control mode	P	S	T	F
Symbol	POT	Default assignment	SI2	I/F circuit	SI	3-30 page		

- Positive direction over-travel inhibit input.
- The operation with this input turned ON is set up in Pr5.04 “Setup of over-travel inhibit input”.
- When using this input, set Pr5.04 “Setup of over-travel inhibit input” to a value other than 1 so that the input is ON when the moving portion of the machine exceeds this signal range toward positive direction.

Title of signal	Negative direction over-travel inhibition input			Related control mode	P	S	T	F
Symbol	NOT	Default assignment	SI1	I/F circuit	SI	3-30 page		

- Negative direction over-travel inhibit input.
- The operation with this input turned ON is set up in Pr5.04 “Setup of over-travel inhibit input”.
- When using this function, set Pr5.04 “Setup of over-travel inhibit input” to a value other than 1 so that the input is ON when the moving portion of the machine exceeds this signal range toward negative direction.

Title of signal	Deviation counter clear input			Related control mode	P	S	T	F
Symbol	CL	Default assignment	SI7	I/F circuit	SI	3-30 page		

- Clears the positional deviation counter.
- Default setup clears the counter at the rising edge of the clear input, To change the setup, modify it in the Pr5.17 “Counter clear input mode”.

<Signal width and clear timing>

Pr5.17	CL signal width	Deviation clear timing
1	500 µs or more	Continually clear the counter while the deviation counter clear input is ON. *1
2	1 ms or more	
3	100 µs or more	Only once clear the counter at OFF to ON edge of the deviation counter clear input signal. *1
4	1 ms or more	

*1 Deviation counter clear input ON/OFF = input photocoupler ON/OFF

Caution This function can be assigned to only SI7. Allocation of this function to any other pin will cause an error.

Related page • P.4-42 “Details of Parameter”

Title of signal	Alarm clear input			Related control mode	P S T F
Symbol	A-CLR	Default assignment	SI8	I/F circuit	SI 3-30 page

- Clears the alarm condition.
- This input cannot clear some alarms.
- For details, refer to P.6-3 When in Trouble “Protective function”, P.2-63 “(8) Alarm Display” and P.7-25 “Display of Battery Alarm”.

Title of signal	Command pulse inhibition input			Related control mode	P S T F
Symbol	INH	Default assignment	SI10	I/F circuit	SI 3-30 page

- Ignores the positional command pulse.
- When using this feature, set Pr5.18 “Invalidation of command pulse inhibition input” to 0.

Caution This function can be assigned to only SI10. Allocation of this function to any other pin will cause an error.

Title of signal	Control mode switching input			Related control mode	P S T F
Symbol	C-MODE	Default assignment	SI9	I/F circuit	SI 3-30 page

- Selects a control mode.

Caution This signal is required in all control modes. No setting will cause an error.

Title of signal	Electronic gear (division/multiplication) switching input 1			Related control mode	P S T F
Symbol	DIV1	Default assignment	SI5	I/F circuit	SI 3-30 page

Title of signal	Electronic gear (division/multiplication) switching input 2			Related control mode	P S T F
Symbol	DIV2	Default assignment	—	I/F circuit	SI 3-30 page

- Up to 4 numerators can be used for command dividing/multiplying by using DIV1 and DIV2.

<DIV1 and DIV2 vs numerator/denominator of selected command dividing/multiplying process>

DIV1	DIV2	Selected command dividing/multiplying process	
		Numerator	Denominator
OFF	OFF	Pr0.09	Pr0.10
ON	OFF	Pr5.00	Pr0.10
OFF	ON	Pr5.01	Pr0.10
ON	ON	Pr5.02	Pr0.10

Title of signal	Damping control switching input 1			Related control mode	P S T F
Symbol	VS-SEL1	Default assignment	SI3	I/F circuit	SI 3-30 page

Title of signal	Damping control switching input 2			Related control mode	P S T F
Symbol	VS-SEL2	Default assignment	—	I/F circuit	SI 3-30 page

- Selects applicable frequency for damping control.
  Combination of damping control input changeover 1 and 2 (VS-SEL1, VS-SEL2) enables select of max. 4 options.

Note Also refer to P.4-21 “Pr2.13 [Damping filter switching selection]”.

Title of signal	Gain switching input			Related control mode	P	S	T	F
Symbol	GAIN	Default assignment	SI4	I/F circuit	SI	3-30 page		

• Select 1st or 2nd gain.

Title of signal	Torque limit switching input			Related control mode	P	S	T	F
Symbol	TL-SEL	Default assignment	—	I/F circuit	SI	3-30 page		

• Select 1st or 2nd torque limit.

Pr5.21	Torque limit switching input (TL-SEL)	Torque limit switching setup (Pr5.23, Pr5.24)	Positive direction Torque limit	Negative direction Torque limit
0			Analog input *1	
1	—	—	Pr0.13	
2	—	—	Pr0.13	Pr5.22
3	OFF	Valid	Pr0.13	
	ON		Pr5.22	
4			Analog input *1	
5				
6	OFF	—	Pr0.13	Pr5.22
	ON		Pr5.25	Pr5.26

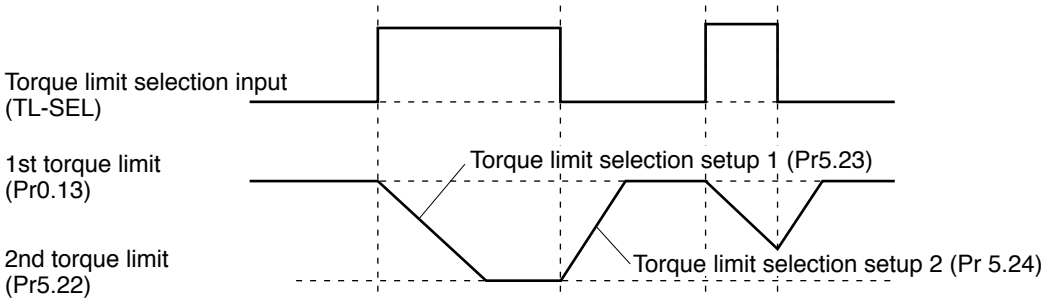
*1 To specify the torque limit value by an analog input, refer to Pr5.21 “Analog torque limit function”.

• Setup of rate of change after torque limit switchover

When applying Pr5.21 “Torque limit selection” = 3, changing rate of torque (slope) after selecting new torque limit can be changed.

When changing from the 1st torque limit to 2nd torque limit, the changing rate (slope) set at Pr5.23 “Torque limit selection setup 1” is applied; after changing from the 2nd torque limit to 1st torque limit, the changing rate (slope) set at Pr5.24 “Torque limit selection setup 2” is applied. The sign of the changing rate is automatically selected by the driver according to the difference in value between the 1st and 2nd torque limit.

If Pr5.23 “Torque limit selection setup 1” and Pr5.24 “Torque limit selection setup 2” are set to 0, switchover is instantaneous.



Caution When the 1st torque limit (Pr0.13) and 2nd torque limit (Pr5.22) are changed from the front panel or through communication, the changing rate setup is ignored and the new torque limit value is immediately and directly applied. That is, changing rate setting is effective only when the selection is made by using the torque limit select input (TL-SEL).

Title of signal	Selection 1 input of internal command speed			Related control mode	P **S** T F
Symbol	**INTSPD1**	Default assignment	**SI10**	I/F circuit	SI 3-30 page

Title of signal	Selection 2 input of internal command speed			Related control mode	P **S** T F
Symbol	**INTSPD2**	Default assignment	**SI7**	I/F circuit	SI 3-30 page

Title of signal	Selection 3 input of internal command speed			Related control mode	P **S** T F
Symbol	**INTSPD3**	Default assignment	**SI5**	I/F circuit	SI 3-30 page

- Select one of 8 internal command speeds.

<Relationship between Pr3.00 "Switching between internal and external speed setup" and internal command speed selection 1-3 and the speed command selected>.

Pr3.00	Selection 1 of internal command speed (INTSPD1)	Selection 2 of internal command speed (INTSPD2)	Selection 3 of internal command speed (INTSPD3)	Selection of speed command
1	OFF	OFF	No effect	1st speed
	ON	OFF		2nd speed
	OFF	ON		3rd speed
	ON	ON		4th speed
2	OFF	OFF	No effect	1st speed
	ON	OFF		2nd speed
	OFF	ON		3rd speed
	ON	ON		Analog speed command
3	The same as Pr3.00=1		OFF	1st to 4th speed
	OFF	OFF	ON	5th speed
	ON	OFF	ON	6th speed
	OFF	ON	ON	7th speed
	ON	ON	ON	8th speed

Title of signal	Speed zero clamp input			Related control mode	P **S** **T** F
Symbol	**ZEROSPD**	Default assignment	**SI3**	I/F circuit	SI 3-30 page

- Set the speed command to 0.
- When using, set Pr3.15 "Speed zero clamp function selection" to a value other than 0.

Title of signal	Speed command sign input			Related control mode	P **S** T F
Symbol	**VC-SIGN**	Default assignment	—	I/F circuit	SI 3-30 page

- Specify the sign of speed command input at velocity control.
  Refer to P.4-24 "Pr3.01 Speed command rotational direction selection"

Title of signal	Torque command sign input			Related control mode	P S **T** F
Symbol	**TC-SIGN**	Default assignment	—	I/F circuit	SI 3-30 page

- Specify the sign of torque command input at torque control.

ON	Negative direction
OFF	Positive direction

Refer to P.4-28 "Pr3.18 Torque command direction selection"

Title of signal	Forced alarm input			Related control mode	P	S	T	F
Symbol	E-STOP	Default assignment	—	I/F circuit	SI	3-30 page		

• Generates Err87.0 “Forced alarm input error”.

Title of signal	Inertia ratio switching input			Related control mode	P	S	T	F
Symbol	J-SEL	Default assignment	—	I/F circuit	SI	3-30 page		

• Selects 1st inertia ratio or 2nd inertia ratio according to the inertia ratio select input (J-SEL).

Inertia ratio switching input (J-SEL)	Applicable inertia ratio
OFF	1st Inertia ratio (Pr0.04)
ON	2nd Inertia ratio (Pr6.12)

Refer to P.4-52 “Pr6.10 Function expansion setup”

Input Signals (Analog Command) and Their Functions

Pin No.	16	Title of signal	Positive direction Torque limit input	Related control mode	P	S	T	F
		Symbol	P-ATL	I/F circuit	AI	3-31 page		

Pin No.	18	Title of signal	Negative direction Torque limit input	Related control mode	P	S	T	F
		Symbol	N-ATL	I/F circuit	AI	3-31 page		

• Specify the torque limit for each direction value by the analog voltage.

Pr5.21	Positive direction Torque limit input (P-ATL)	Negative direction Torque limit input (N-ATL)	Positive direction Torque limit	Negative direction Torque limit
0	0 to 10V	–10 to 0V	P-ATL	N-ATL
1	—		Set up through parameter. *1	
2				
3				
4	0 to 10V	0 to 10V	P-ATL	N-ATL
5	0 to 10V	No effect	P-ATL	
6	—		Setup by parameter *1	

*1 When specifying the torque limit value through the parameter, refer to P.4-47 “Torque limit select function”

Pin No.	14	Title of signal	Speed command input	Related control mode	P **S** T F
		Symbol	SPR	I/F circuit	AI 3-31 page

- Input the speed command in the form of analog voltage.
- The table below shows relationship between the combination of Pr3.00 “Switching between internal and external speed setup”, Pr3.01 “Speed command direction selection”, Pr3.03 “Speed command input inversion”, analog speed command (SPR) of I/F connector and speed command sign selection (VC-SIGN) and the motor rotational direction; and the conversion graph of analog speed command input voltage to the speed command.

Pr3.00	Pr3.01	Pr3.03	Speed command input (SPR)	Speed command sign selection (VC-SIGN)	Motor rotational direction
0	0	0	+Voltage (0 to 10V)	No effect	Positive direction
			–Voltage (–10 to 0V)	No effect	Negative direction
		1	+Voltage (0 to 10V)	No effect	Negative direction
			–Voltage (–10 to 0V)	No effect	Positive direction
	1	No effect	+Voltage (0 to 10V)	OFF	Positive direction
			–Voltage (–10 to 0V)		
			+Voltage (0 to 10V)	ON	Negative direction
			–Voltage (–10 to 0V)		

Pin No.	14 16	Title of signal	Torque command input	Related control mode	P S **T** F
		Symbol	TRQR	I/F circuit	AI 3-31 page

- Input the torque command in the form of analog voltage.
  When Pr3.19 “Torque command selection” = 0, pin No.14
  When Pr3.19 “Torque command selection” = 1, pin No.16

Pr3.17	Pr3.18	Pr3.20	Torque command input (TRQR)	Torque command sign selection (TC-SIGN)	Motor rotational direction
0	0	0	+Voltage (0 to 10V)	No effect	Positive direction
			–Voltage (–10 to 0V)	No effect	Negative direction
		1	+Voltage (0 to 10V)	No effect	Negative direction
			–Voltage (–10 to 0V)	No effect	Positive direction
	1	No effect	+Voltage (0 to 10V)	OFF	Positive direction
			–Voltage (–10 to 0V)		
			+Voltage (0 to 10V)	ON	Negative direction
			–Voltage (–10 to 0V)		

Pin No.	14	Title of signal	Speed limit input	Related control mode	P S **T** F
		Symbol	SPL	I/F circuit	AI 3-31 page

- When setting Pr3.17 “Torque command selection” to 1, input the speed limit value in the form of analog voltage.

4. Inputs and outputs on connector X4

Output Signal and Pin No.

Output Signals (Common) and Their Functions

Control output signal of desired function can be assigned to I/F connector. Logic of the output pin cannot be changed.

				Applicable parameter	Default parameter setting ( ): decimal notation	Default Setup		
						Position/ Full-closed control	Verocity control	Torque control
						Signal	Signal	Signal
Pin No.	10 11	Title of signal	**SO1 output**	Pr4.10	00030303h (197379)	BRK-OFF	BRK-OFF	BRK-OFF
		Symbol	Pin No.10: **SO1–** Pin No.11: **SO1+**					
Pin No.	34 35	Title of signal	**SO2 output**	Pr4.11	00020202h (131586)	S-RDY	S-RDY	S-RDY
		Symbol	Pin No.34: **SO2–** Pin No.35: **SO2+**					
Pin No.	36 37	Title of signal	**SO3 output** *1	Pr4.12	00010101h (65793)	ALM	ALM	ALM
		Symbol	Pin No.36: **SO3–** Pin No.37: **SO3+**					
Pin No.	38 39	Title of signal	**SO4 output**	Pr4.13	00050504h (328964)	INP	AT-SPEED	AT-SPEED
		Symbol	Pin No.38: **SO4–** Pin No.39: **SO4+**					
Pin No.	12	Title of signal	**SO5 output**	Pr4.14	00070707h (460551)	ZSP	ZSP	ZSP
		Symbol	**SO5**					
Pin No.	40	Title of signal	**SO6 output**	Pr4.15	00060606h (394758)	TLC	TLC	TLC
		Symbol	**SO6**					

- The function is changed by the setting of parameter.
  See “Functions assignable to control output” as shown below.

Note *1 Fixed ALM output.
[—]: No function assigned

Function allocatable to control input

Title of signal	**Servo-Alarm output**			Related control mode	P	S	T	F
Symbol	**ALM**	Default assignment	**SO3**	I/F circuit	SO1	3-32 page		

- This signal shows that the driver is in alarm status..
- Output transistor turns ON when the driver is at normal status, and turns OFF at alarm status.

Title of signal	**Servo-Ready output**			Related control mode	P	S	T	F
Symbol	**S-RDY**	Default assignment	**SO2**	I/F circuit	SO1	3-32 page		

- This signal shows that the driver is ready to be activated.
- Output transistor turns ON when both control and main power are ON but not at alarm status. When in the absolute mode and absolute I/F function is enabled, turns on the output transistor upon completion of the absolute data transfer provided that the conditions shown above are met.

Title of signal	External brake release signal			Related control mode	P	S	T	F
Symbol	BRK-OFF	Default assignment	SO1	I/F circuit	SO1	3-32 page		

- Feeds out the timing signal which activates the electromagnetic brake of the motor.
- Turns the output transistor ON at the release timing of the electro-magnetic brake.

Title of signal	Positioning complete			Related control mode	P	S	T	F
Symbol	INP	Default assignment	SO4	I/F circuit	SO1	3-32 page		

Title of signal	Positioning complete 2			Related control mode	P	S	T	F
Symbol	INP2	Default assignment	—	I/F circuit	SO1	3-32 page		

- Outputs the positioning complete signal/positioning complete signal 2.
- Turns ON the output transistor upon completion of positioning.

Title of signal	Speed arrival output			Related control mode	P	S	T	F
Symbol	AT-SPPED	Default assignment	SO4	I/F circuit	SO1	3-32 page		

- Outputs thespeed arrival signal.
- Turns ON the output transistor upon arrive of speed.

Title of signal	Torque in-limit signal output			Related control mode	P	S	T	F
Symbol	TLC	Default assignment	SO6	I/F circuit	SO2	3-32 page		

- Outputs thetorque in-limit signal.
- Turns ON the output transistor upon limit of torque.

Title of signal	Zero-speed detection output signal			Related control mode	P	S	T	F
Symbol	ZSP	Default assignment	SO5	I/F circuit	SO2	3-32 page		

- Outputs the zero-speed detection signal.
- Turns ON the output transistor upon detection of Zero-speed.

Title of signal	Speed coincidence output			Related control mode	P	S	T	F
Symbol	V-COIN	Default assignment	—	I/F circuit	SO1	3-32 page		

- Outputs the speed coincidence signal.
- Turns ON the output transistor upon coincidence of speed.

Title of signal	Alarm output 1			Related control mode	P	S	T	F
Symbol	WARN1	Default assignment	—	I/F circuit	SO1	3-32 page		

- Outputs the warning output signal set to Pr4.40 “Warning output select 1”.
- Turns ON the output transistor upon occurrence of warning condition.

Title of signal	Alarm output 2			Related control mode	P	S	T	F
Symbol	WARN2	Default assignment	—	I/F circuit	SO1	3-32 page		

- Outputs the warning output signal set to Pr4.41 “Warning output select 2”.
- Turns ON the output transistor upon occurrence of warning condition.

• Selection of alarm 1 output and 2 output

Alarm No.	Alarm	Content	Pr6.27 *1	Pr4.40/ Pr4.41 *2	Pr6.38 Corresponding bit *3
A0	Overload protection	Load factor is 85% or more the protection level.	○	1	bit7
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.	○	2	bit5
A2	Battery alarm	Battery voltage is 3.2 V or lower.	Fixed at no time limit.	3	bit0
A3	Fan alarm	Fan has stopped for 1 sec.	○	4	bit6
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.	○	5	bit4
A5	Encoder overheat alarm	The encoder detects overheat alarm.	○	6	bit3
A6	Oscillation detection alarm	Oscillation or vibration is detected.	○	7	bit9
A7	Lifetime detection alarm	The life expectancy of capacity or fan becomes shorter than the specified time.	Fixed at no time limit.	8	bit2
A8	External scale error alarm	The feedback scale detects the alarm.	○	9	bit8
A9	External scale communication alarm	The number of successive feedback scale communication errors exceeds the specified value.	○	10	bit10

*1 The “circle” means that a time in the range 1 to 10s or no time limit can be selected through Pr6.27 “Warning latching time”. Note that the battery warning and the end of life warning have no time limit.

*2 Select the warning output signal 1 (WARN1) or warning output signal 2 (WARN2) through Pr4.40 “Warning output select 1” or Pr4.41 “Warning output select 2”. When the set value is 0, all warnings are ORed before being output. Do not set to any value other than those specified in the table above.

*3 A warning detection can be masked by Pr6.38 “Warning mask setup” Corresponding bits are shown in the table. Warning is masked with bit = 1.

Title of signal	Positional command ON/OFF output			Related control mode	P	S	T	F
Symbol	P-CMD	Default assignment	—	I/F circuit	SO1	3-32 page		

- Turns on output transistor with positional command applied.

Title of signal	Speed in-limit output			Related control mode	P	S	T	F
Symbol	V-LIMIT	Default assignment	—	I/F circuit	SO1	3-32 page		

- Turns on output transistor when the speed is limited by torque controlling function.

Title of signal	Alarm attribute output			Related control mode	P	S	T	F
Symbol	ALM-ATB	Default assignment	—	I/F circuit	SO1	3-32 page		

- Turns on output transistor when an alarm that can be cleared generates.

Title of signal	Speed command ON/OFF output			Related control mode	P	S	T	F
Symbol	V-CMD	Default assignment	—	I/F circuit	SO1	3-32 page		

• Turns on output transistor when the speed command is applied while the speed is controlled.

Output Signals (Pulse Train) and Their Functions

Pin No.	21 / 22	Title of signal	A-phase output	Related control mode	P	S	T	F
		Symbol	Pin No.21: **OA+** Pin No.22: **OA–**	I/F circuit	PO1	3-32 page		

Pin No.	48 / 49	Title of signal	B-phase output	Related control mode	P	S	T	F
		Symbol	Pin No.48: **OB+** Pin No.49: **OB–**	I/F circuit	PO1	3-32 page		

Pin No.	23 / 24	Title of signal	Z-phase output	Related control mode	P	S	T	F
		Symbol	Pin No.23: **OZ+** Pin No.24: **OZ–**	I/F circuit	PO1	3-32 page		

• Feeds out the divided encoder signal or feedback scale signal (A, B, Z-phase) in differential. (equivalent to RS422)
• Ground for line driver of output circuit is connected to signal ground (GND) and is not insulated.
• Max. output frequency is 4Mpps (after quadrupled)

Pin No.	19	Title of signal	Z-phase output	Related control mode	P	S	T	F
		Symbol	**CZ**	I/F circuit	PO1	3-32 page		

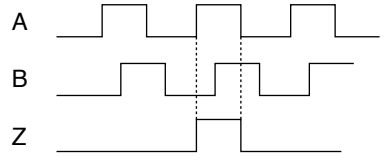
• Open collector output of Z-phase signal
• The emitter side of the transistor of the output circuit is connected to the signal ground (GND) and is not insulated.

Note

• **When the output source is the encoder**

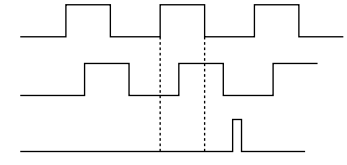
• If the encoder resolution × $\frac{\text{Pr0.11}}{\text{Pr5.03}}$ is multiple of 4, Z-phase will be fed out synchronizing with A-phase. In other case, the Z-phase width will be equal to the encoder resolution, and will not synchronize with A-phase because of narrower width than that of A-phase.

when the encoder resolution × $\frac{\text{Pr0.11}}{\text{Pr5.03}}$ is multiple of 4,



synchronized

when the encoder resolution × $\frac{\text{Pr0.11}}{\text{Pr5.03}}$ is not multiple of 4,



not-synchronized

• In case of the 5-wire, 20-bit incremental encoder, the signal sequence might not follow the above fig. until the first Z-phase is fed out. When you use the pulse output as the control signal, rotate the motor one revolution or more to make sure that the Z-phase is fed out at least once before using.

Output Signals (Analog) and Their Functions

Pin No.	42	Title of signal	Torque monitor output	Related control mode	P	S	T	F
		Symbol	IM	I/F circuit	AO	3-33 page		

- Definition of the output signal varies with the output of Pr4.18 (analog monitor 2 type).
- The output signal is identical to the analog monitor 2 on the front monitor.

Pin No.	43	Title of signal	Speed monitor output	Related control mode	P	S	T	F
		Symbol	SP	I/F circuit	AO	3-33 page		

- Definition of the output signal varies with the output of Pr4.16 (analog monitor 1 type).
- The output signal is identical to the analog monitor 1 on the front monitor.

Output Signals (Others) and Their Functions

Pin No.	13, 15 17, 25	Title of signal	**Signal ground**	Related control mode	P	S	T	F
		Symbol	**GND**	I/F circuit	—			

- Signal ground
- This output is insulated from the control signal power (COM–) inside of the driver.

Pin No.	50	Title of signal	**Frame ground**	Related control mode	P	S	T	F
		Symbol	**FG**	I/F circuit	—			

- This output is connected to the earth terminal inside of the driver.